BUDGETBITE

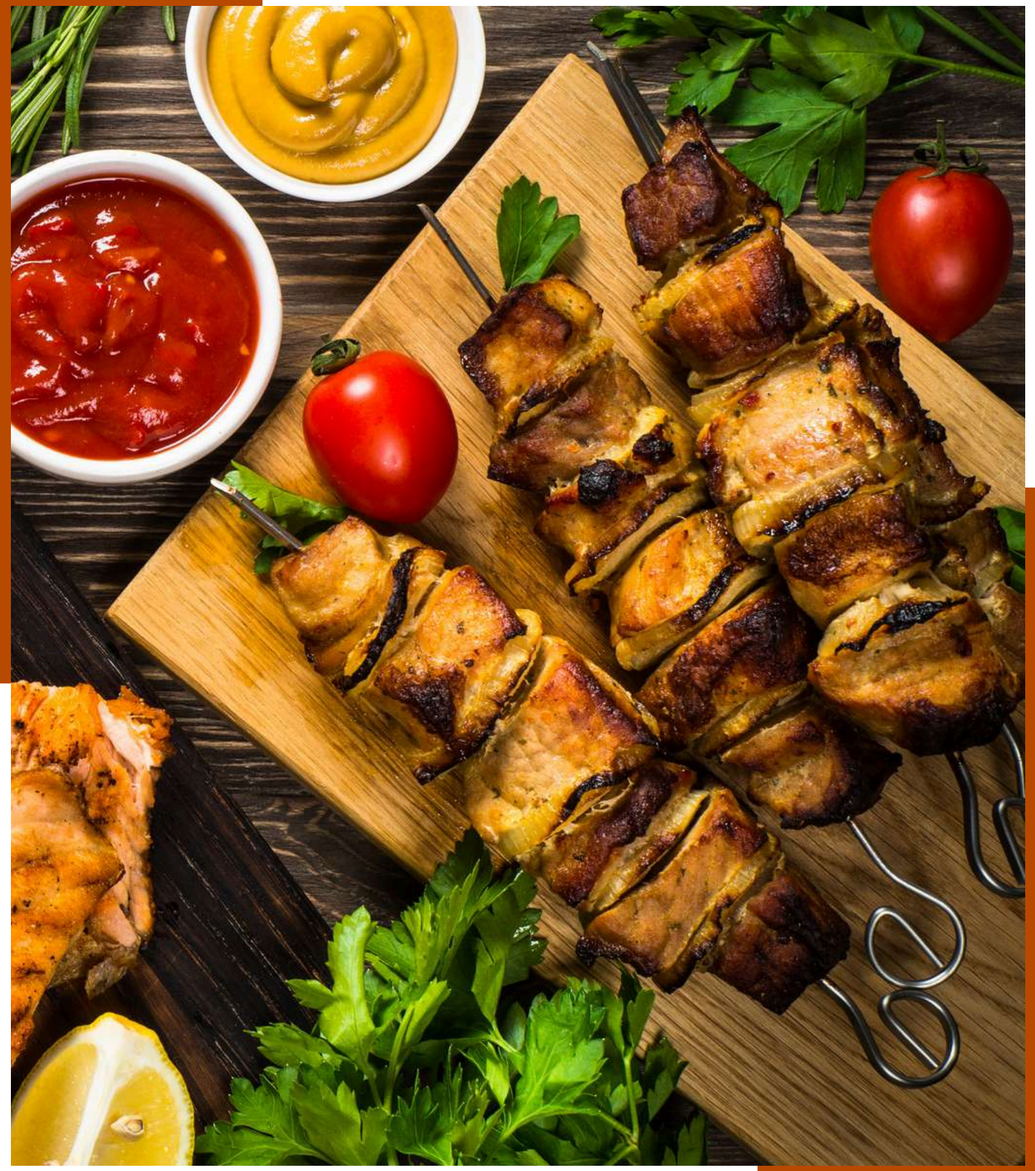
Presentation

About BudgetBite

BudgetBite is a AI powered meal generator with recipe reference and price estimation.

Why BudgetBite ?

BudgetBite was inspired by a laziness to figure out what to cook and the ability to budget recipes so that people who are financially conscious can make a good meal.



Backend
using Flask

we used flask for our backend because of its simplicity in integration with react native pwa and the ability to write code in python keeping simple.

Frontend

PWA React Native

We used React Native and Expo Go for the ability to create a progressive web app meaning this can be a website, iphone app, and adroid app. This integration with React Native made it simple and achievable in our time frame.

Gemini API

we initially were going to use a computer vision system but we quickly realized that would be a bottleneck because of our time constraints so what we pivoted to was passing an image to the gemini api instead and this works great for testing and development. This allows us to show a working demo without spending weeks collecting data and labeling said data.

Database
SQlite

We use SQLite to hold recipes this is meant to serve two purposed to have common recipes and instructions to be used instead of the heavy gemini recipe generation.

- Name of Recipe
- Ingredients
- Instructions
- Image
- Estimated Price

Kroger API

We used the Kroger API to give an estimate of the ingredients and for budget tiers. If the recipe exceeds the selected budget we prompt the user for regeneration.

Ground Truth Testing

We used online images and labeled them manually and tested them with the Gemini API while recall was high meaning Gemini was able to detect 90% of the ingrediants in the image but precision was low meaning that even though it was able to detect the ingrediants it had many ingrediants that were not in the photo. this gave us an overall f1 score of 60%

What does that mean?

???

A CV model will be needed for production
because of the low f1 score we cannot trust gemini alone in production

Longer development time
data agregation and data labeling is not a fast process. This process can take many months to complete

Good to demo
on the positive because of the high recall gemini is able to deliver a strong number of correct ingrediants allowing the recipe generation to be accurate

recipe generation uses Gemini
We use gemini to generate recipes as well as finding the ingredients

Valid/Refusal Flow

we created a valid and refusal flow architecture that allows the recipe generationg to be deinied or accepted based on the ingrediants extracted

Individual Contributions

Demo

THANK YOU

for listening to our Presentation

ANY QUETIONS?